20. If MR and MC denote the marginal revenue and marginal cost and $MR - MC = 36x - 3x^2 - 81$, then the maximum profit at x is equal to

(a) 3 (b) 6 (c) 9 (d)5

21. If the marginal revenue of a firm is constant, then the demand function is

(a) MR (b) MC (c) $C(x)$ (d) AC

22. For a demand function p, if $\int \frac{dp}{p} = k\int \frac{dx}{x}$ then k is equal to

(a) η_d (b) $-\eta_d$ (c) $\frac{-1}{\eta_d}$ (d) $\frac{1}{\eta_d}$

23. Area bounded by $y = e^x$ between the limits 0 to 1 is

(a)$(e-1)$ sq.units (b)$(e+1)$ sq.units

(c)$\left(1-\frac{1}{e}\right)$ sq.units (d) $\left(1+\frac{1}{e}\right)$ sq.units

24. The area bounded by the parabola $y^2 = 4x$ bounded by its latus rectum is

(a) $\frac{16}{3}$ *sq.units* (b) $\frac{8}{3}$ *sq.units*

(c) $\frac{72}{3}$ *sq.units* (d) $\frac{1}{3}$ *sq.units*

25. Area bounded by $y = |x|$ between the limits 0 and 2 is

(a) 1sq.units (b) 3 sq.units

(c) 2 sq.units (d) 4 sq.units

Miscellaneous problems

1. A manufacture's marginal revenue function is given by MR= $275 - x - 0.3x^2$. Find the increase in the manufactures total revenue if the production is increased from 10 to 20 units.

2. A company has determined that marginal cost function for x product of a particular commodity is given by $MC = 125 + 10x - \frac{x^2}{9}$. Where C is the cost of producing x units of the commodity. If the fixed cost is ₹ 250 what is cost of producing 15 units

3. The marginal revenue function for a firm is given by $MR = \frac{2}{x+3} - \frac{2x}{(x+3)^2} + 5$. Show that the demand function is $P = \frac{2}{x+3} + 5$.

4. For the marginal revenue function $MR = 6 - 3x^2 - x^3$, Find the revenue function and demand function.

5. The marginal cost of production of a firm is given by $C'(x) = 20 + \frac{x}{20}$ the marginal revenue is given by $R'(x) = 30$ and the fixed cost is ₹ 100. Find the profit function.

6. The demand equation for a product is $p_d = 20 - 5x$ and the supply equation is $p_s = 4x + 8$. Determine the consumer's surplus and producer's surplus under market equilibrium.

7. A company requires $f(x)$ number of hours to produce 500 units. It is represented by $f(x) = 1800x^{-0.4}$. Find out the number of hours required to produce additional 400 units. [$(900)^{0.6}$=59.22, $(500)^{0.6}$=41.63]

8. The price elasticity of demand for a commodity is $\frac{p}{x^3}$. Find the demand function if the quantity of demand is 3, when the price is ₹2.

9. Find the area of the region bounded by the curve between the parabola $y = 8x^2 - 4x + 6$ the y-axis and the ordinate at x = 2.

10. Find the area of the region bounded by the curve $y^2 = 27x^3$ and the lines x = 0, y = 1 and y = 2.

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